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A fully distributed event-Triggered control approach to leader-following consensus of multiagent systems

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Abstract

This paper investigates the leader-following consensus problem for general linear multiagent systems under a fully distributed manner. To avoid using the global information of the communication network, two node-based adaptive parameters are designed and introduced to the fully distributed event-based control law. Such a protocol is more scalable and flexible for large-scaled and complex networked system, such as microgrids or multiple unmanned vehicles. A novel event-triggered rule is presented, and an internal dynamic variable is introduced in it to further cut down the triggering times and save the communication resources. A well-designed Lyapunov function candidate is constructed to not only ensure the convergence of the close-looped systems but also avoid using the eigenvalue of Laplacian matrix in the dynamic variable. Moreover, it is proved that the designed event-triggered conditions can avoid the Zeno behaviour. Finally, a simulation example based on a group of unmanned intelligent vehicle systems is provided to demonstrate the validity of the designed control method.

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1. Introduction

Coordination of multiagent systems is undoubtedly a hot spot in the field of automation over the past two decades. Consensus problem is one of the most fundamental and vital problems of multiagent systems. To resolve the consensus problem, a distributed protocol is always designed to make the states of agents reach the same desired value. The potential applications of the consensus control in wireless sensor networks, smart grids, spacecraft and other real engineering systems can be found in [1–4]. Many pioneer works about consensus problem have been reported in [5–9].

Different from the traditional leaderless consensus problem, the leader-following consensus is another hot topic. The leaderless consensus requires that the states of all agents reach the same value, while leader-following consensus further requires that this value be the same as the state of the leader. And thus such a problem can also be regarded as a tracking control problem. In [10], the leader-following consensus of general linear multiagent systems under both fixed and switching topologies is studied and the dynamic random graphs is further studied in [11]. The leader-following consensus with state constraints is studied in [12]. The leader-following consensus for a kind of nonlinear multiagent systems is investigated in [13], where a composite fuzzy distributed protocol is developed. In [14], the observer and stochastic faulty actuator-based leader-following consensus protocol has been provided. The stochastic faulty estimator-based non-fragile tracking controller is further studied in [15]. Moreover, the potential applications in practical systems for leader-following consensus are also widely reported in literature. For example, the leader-following consensus is combined with trajectory planning algorithm in [16] to address the parafoil airdrop problem. And the fixed-time leader-following consensus for the multiagent systems consisting of several wheeled mobile robots is studied in [17].

Due to the wide applications of embedded microprocessors, the digitized discrete-time controller attracts much attention. To resolve the consensus problem in such a scenario, some time-based control protocols are designed in [18,19]. In those works, how to select an appropriate sampling period such that not only the communication resources can be retrenched but also the system performance can be ensured is a very difficult question. To ensure the stability, the sampling period is often selected very conservatively. This means that although when the system states change slowly, the controller is still updated according to the fixed sampling period, which leads to a wastefulness of the communication bandwidth. To overcome this drawback, an event-based mechanism is firstly designed in [20] to save the communication bandwidth and prolong the life-span of the whole multiagent system. In such a case, the controller will not be updated continuously or periodically, and will be updated by the violation of a predefined event. In [21], the event-triggered consensus problem for the general linear multiagent systems under undirected network is studied. In [22], an event-based controller is designed for multiagent systems with sensor faults and input saturation. Moreover, a dynamic event-triggered approach is adequately studied in [23], where a dynamic event-triggered control framework for the studied system is developed.

The most of consensus control protocols mentioned above are designed for single agent, and the protocols only rely on the information of the agent and its neighbours. Therefore, this class of controllers are called distributed controllers. However, in those works, some global information is required to be known beforehand during the controller designing process. For example, the minimum non-negative eigenvalue of the Laplacian matrix is used in [24]. Moreover, the total number of the studied multiagent system is required in [25] to solve the controller gain and further determine the triggering function. Thus, the proposed control laws in those papers are distributed but not fully distributed. For a real engineering multiagent system, the communication network among agents may be huge, complex and time-varied. In this case, the above mentioned protocol is difficult to be implemented because the global information is difficult to obtain and has to be reacquired when the topology is changed. To resolve this issue and develop a scalable controller, a fully distributed strategy is presented in [26]. An edge-based adaptive coupling weight is designed and introduced to the triggering function so that the global information is no longer needed. In [27], a node-based adaptive parameter is introduced to resolve the fully distributed consensus for leaderless multiagent systems. The absolute information of agents is no longer needed in this article, and only the sampled relative information is used in the controller. Moreover, bipartite consensus control for the multiagent systems is developed in [28] via a fully distributed approach. More recent results on fully distributed mechanism are reported in [29–32].

Though some event-based fully distributed results have been reported in literature, most of them consider the leaderless multiagent systems. For the leader-following multiagent systems, we should not only consider how to achieve the consensus among followers, but also enable them to follow the leader. Thus, a more complex control framework has to be considered in our work compared to the leaderless case. For event-triggered control mechanism, obtaining a larger minimum inter-execution time between two consecutive triggering instants is another interesting problem. In order to enlarge such an inter-execution interval, some novel triggering mechanisms such as dynamic event-triggered control, Lyapunov-function-based event-triggered control are provided. In [33] and [34], some early results have proved that the average inter-execution interval of dynamic event-triggered mechanism can be observably amplified due to the introduction of non-negative internal dynamic variable. However, these obtained results are also global information-dependent. How to degrade the waste of the bandwidth in multiagent systems while avoiding the usage of global information needs further investigation.

Motivated by above discussion, this paper intends to address the leader-following consensus problem of multiagent systems via a fully distributed dynamic event-triggered control. The contributions of this paper are generalized as follows.

1. A fully distributed and scalable protocol for the leader-following multiagent systems is designed under the event-triggered fashion. Neither the information of Laplacian matrix nor the number of agents is required in such a design, and thus the proposed strategy is more applicable in practical engineering. Estimators are developed for each agent to get their estimated state information to avoid continuous communications in either controller updates or triggering conditions monitoring. Two novel node-based adaptive parameters are developed for the agents that connected to the leader or not, respectively.
2. A dynamic triggering rule is designed in this article to determine when the controller and estimators update. Different from the static one, a positive dynamic internal variable is introduced in the triggering function to further cut down the triggering times and save communication resources. Compared with some previous results in [35–37], more complex leader-following multiagent systems are considered in our work. It is quite difficult to ensure the stability when the update frequency of the controller is reduced, and it is more difficult when a leader-following structure is considered.
3. One of the coefficient in the dynamic variable has to be initialized by some global information if the traditional Lyapunov function that presented in [36–38] is used. Thus, a new well-designed Lyapunov function candidate is provided in this article to relax this

limitation, based on which, it is proved that the leader-following consensus can be reached. Moreover, the Zeno behaviour can be avoided and the minimum inter-event time is also provided. Finally, a simulation is conducted to demonstrate the validity of designed controller.

Some preliminaries and problem formulation are given in Section 2. The fully distributed dynamic event-triggered mechanism design for the studied multiagent system is given in Section 3. A simulation example is conducted to illustrate the validity of the obtained results in Section 4. Finally, Section 5 concludes this paper.

Notation. $\lambda_i(Q) and \lambda_{\min}(Q)$ denote the i-th and minimum eigenvalues of a non-singular matrix Q, respectively. $\mathbf{1}_N$ is a vector with entries being one. $\mathrm{diag}[d_1, \cdots, d_N]$ denotes a diagonal matrix with d_i being its diagonal entries. The Kronecker production of two matrices M, N is denoted as $M \otimes N$.

2. Preliminaries and problem statement

In the paper, we use $\mathsf{G} = \{\mathcal{Y}, \mathcal{Z}\}$ to denote an undirected graph with $\mathcal{Y} = \{1, 2, \ldots, N\}$ and $\mathcal{Z} \subseteq \mathcal{Y} \times \mathcal{Y}$ being the agent set and edge set, respectively. Let $\mathcal{N}_p = \{q \in \mathcal{Y} : \{p, q\} \in \mathcal{Z}, p \neq q\}$ denote the neighboring set of agent p. Adjacency matrix of G is defined as $\mathcal{A} = [a_{pq}] \in \mathbb{R}^{N\times N}$, where $a_{pp} = 0$, $a_{pq} = 1$, iff $\{p, q\} \in \mathcal{Z}$ and $a_{pq} = 0$ otherwise. We also define a diagonal matrix $\mathcal{D} = \mathrm{diag}[d'_1, \cdots, d'_N]$ with the diagonal element being $d'_p = \sum_{l\in\mathcal{N}_p} a_{pq}$. Then, the Laplacian matrix can be defined as $L = \mathcal{D} - \mathcal{A}$. The topology consisting of N agents and agent 0 can be described by a graph $\bar{\mathsf{G}}$. Another diagonal matrix $D = \mathrm{diag}[d_1, \cdots, d_N]$, where $d_p = 1$ if agent p can use the leader's information and $d_p = 0$ otherwise.

In this work, a leader-following multiagent system is considered. The leader's dynamics is

$$\dot{x}_0(t) = Ax_0(t), \tag{1}$$

where $x_0(t) \in \mathbb{R}^n$ is the state of leader and A is the system matrix with compatible dimensions. The followers' dynamics is given as

$$\dot{x}_p(t) = Ax_p(t) + Bu_p(t), \forall p \in \mathcal{Y} \tag{2}$$

where $x_p(t)$ is the state of agent p and it has the same dimension of the leader; $u_p(t) \in \mathbb{R}^m$ denotes the control law to be designed of agent p; B is the input matrix of the followers with compatible dimension.

Now, we present the formal definition of leader-following consensus which is the same as [38].

Definition 1 ([38]). For a designed event-based control protocol $u_p(t)$, the leader-following consensus problem of system (1)-(2) is resolved successfully as long as

$$\lim_{t\to\infty} \|x_p(t) - x_0(t)\| = 0, \forall p \in \mathcal{Y}. \tag{3}$$

As shown in [38], to address the leader-following consensus, the protocol is usually designed as

$$u_p(t) = K\left[\sum_{q\in\mathcal{N}_p} a_{pq}(x_q(t) - x_p(t)) + d_p(x_0(t) - x_p(t))\right] \tag{4}$$

for all $p \in \mathcal{Y}$. Here, K is the controller gain and usually chosen as $K = B^T\mathbb{F}$ with $\mathbb{F}$ being the solution to the algebraic Riccati equation (ARE) $A^T\mathbb{F} + \mathbb{F}A - c_0\mathbb{F}BB^T\mathbb{F} + Q = 0$, where c_0 is a positive scalar related to the eigenvalue of an extended Laplacian matrix, and Q can be any positive definite matrix.

Note that in the above design procedure, because of the presence of c_0, global information of the topology graph has to be known beforehand. In addition, the state of each agent p as well as its neighbours is required in the above control protocol (4). Thus, continuous communication among leader and followers is also required. The aim of this article is to develop an event-based mechanism so that the requirement of continuous communication as well as global information are removed.

We are supposed to design an event-based fully distributed control input $u_p(t)$ so that the consensus is reached, and the event-triggered rule is Zeno-free. Moreover, the paper bases on the following two standard assumptions.

Assumption 1. The matrix pair (A, B) is stabilizable.

Assumption 2. The topology graph $\bar{\mathsf{G}}$ is connected.

3. Fully distributed dynamic event-triggered mechanism

To achieve the leader-following consensus, the event-triggered control input is given as

$$\begin{aligned} u_p(t) =& K w_p(t), \\ w_p(t) =& \mu_p(t) \sum_{q \in \mathcal{N}_p} a_{pq}(\tilde{x}_q(t) - \tilde{x}_p(t)) + \nu_p(t) d_p (x_0(t) - \tilde{x}_p(t)), \end{aligned} \tag{5}$$

where $\tilde{x}_p(t), p \in \mathcal{Y}$ is the estimation value of the agent state with $\tilde{x}_p(t) = e^{A(t - t_s^p)} x_p(t_s^p)$, t_s^p is the s-th triggering instant for agent p. $\mu_p(t)$ and $\nu_p(t)$ are two node-based adaptive parameters which will be defined later, and the controller gain K can be acquired by solving the following ARE

$$A^T\mathbb{F} + \mathbb{F}A - \mathbb{F}BB^T\mathbb{F} + Q = 0. \tag{6}$$

Here, Q is a positive definite matrix, and $\mathbb{F} > 0$ is a feasible solution of (6). K can be given as $K = B^T\mathbb{F}$.

Before going any further, we define two error variables, with

$$e_p(t) = \tilde{x}_p(t) - x_p(t), t \in [t_s^p, t_{s+1}^p), \tag{7}$$

and

$$\psi_p(t) = x_p(t) - x_0(t), \tag{8}$$

which are the estimation error and consensus error, respectively. For easy of illustration, we define $\psi(t) = [\psi_1^T(t), \cdots, \psi_N^T(t)]^T$ and $w(t) = [w_1^T(t), \cdots, w_N^T(t)]^T$. We also define $\mu(t) = \mathrm{diag}[\mu_1(t), \cdots, \mu_N(t)]$ and $\nu(t) = \mathrm{diag}[\nu_1(t), \cdots, \nu_N(t)]$. After that, we can derive from the definition of $w_p(t)$ that

$$w(t) = -[(\mu L + \nu D) \otimes I_n]\tilde{x}(t) + (\nu D \otimes I_n)(\mathbf{1}_N \otimes x_0(t)). \tag{9}$$

By using $\psi_p(t) = \tilde{x}_p(t) - e_p(t) - x_0(t)$, we have that

$$\psi(t) = \tilde{x}(t) - e(t) - \mathbf{1}_N \otimes x_0(t), \tag{10}$$

where $e(t) = [e_1^T(t), \cdots, e_N^T(t)]^T$. Combining the above equations (9) and (10), we have

$$w(t) = -(\mu L \otimes I_n)\tilde{x}(t) - (\nu D \otimes I_n)(e(t) + \psi(t)). \tag{11}$$

It is also clear that for any $t \in [t_s^p, t_{s+1}^p]$, the derivative of $e_p(t)$ is

$$\dot{e}_p(t) = Ae_p(t) - BKw_p(t), \forall p \in \mathcal{Y}. \tag{12}$$

Combining (10) and (12), we can obtain the time derivative of $\psi(t)$ as

$$\dot{\psi}(t) =(I_N \otimes A - \nu D \otimes BK)\psi(t) - (\nu D \otimes BK)e(t) - (\mu L \otimes BK)\tilde{x}(t).$$

For easy reading, the time t will be omitted hereafter unless there is a specification. Define

$$\tilde{x}_p - \tilde{x}_q = \tilde{\psi}_p - \tilde{\psi}_q \triangleq \tilde{z}_{pq}, \{p, q\} \in \mathcal{Z}. \tag{13}$$

with $\tilde{\psi}_p = \tilde{x}_p - x_0$.

In this paper, the following triggering rule is given

$$t_{s+1}^p \triangleq \inf \left\{ t > t_s^p \middle| f_p(t) \geq 0 \right\}. \tag{14}$$

The triggering function is designed as follows

$$f_p(t) =\sigma_p\left\{\frac{1}{2}(1+\epsilon\nu_p)d_p e_p^T\Gamma e_p + \sum_{q=1}^{N}(1+\epsilon\mu_p)a_{pq}e_p^T\Gamma e_p - \frac{1}{4}\sum_{q=1}^{N}a_{pq}\tilde{z}_{pq}^T\Gamma\tilde{z}_{pq} - \frac{1}{2}d_p\tilde{\psi}_p^T\Gamma\tilde{\psi}_p\right\} - \delta_p(t), \tag{15}$$

where σ_p and ϵ are positive scalars, $\Gamma = K^T K$, μ_p and ν_p are two adaptive parameters satisfying

$$\dot{\mu}_p(t) =k_p\sum_{q=1}^{N}a_{pq}\tilde{z}_{pq}^T K^T K\tilde{z}_{pq}, \tag{16}$$

$$\dot{\nu}_p(t) =k_p d_p\tilde{\psi}_p^T K^T K\tilde{\psi}_p, \tag{17}$$

where k_p is a positive constant and the initial values of $\mu(t)$ and $\nu(t)$ can be selected such that $\mu(t_0) \geq 0$ and $\nu(t_0) \geq 0$.

Here, $\delta_p(t)$ is introduced as an internal variable to enlarge the inter-execution time interval. Such a function is given as

$$\dot{\delta}_p(t) = -\left\{\frac{1}{2}(1+\epsilon\nu_p)d_p e_p^T\Gamma e_p + \sum_{q=1}^{N}(1+\epsilon\mu_p)a_{pq}e_p^T\Gamma e_p - \frac{1}{4}\sum_{q=1}^{N}a_{pq}\tilde{z}_{pq}^T\Gamma\tilde{z}_{pq} - \frac{1}{2}d_p\tilde{\psi}_p^T\Gamma\tilde{\psi}_p\right\} - \eta\delta_p(t), \tag{18}$$

where $\delta_p(0) > 0$ and η is a positive constant.

Clearly, according to (14) and (15), we can check that the following inequality holds for all $t \in [0, \infty)$:

$$\sigma_p\left\{\frac{1}{2}(1+\epsilon\nu_p)d_p e_p^T\Gamma e_p + \sum_{q=1}^{N}(1+\epsilon\mu_p)a_{pq}e_p^T\Gamma e_p - \frac{1}{4}\sum_{q=1}^{N}a_{pq}\tilde{z}_{pq}^T\Gamma\tilde{z}_{pq} - \frac{1}{2}d_p\tilde{\psi}_p^T\Gamma\tilde{\psi}_p\right\} - \delta_p(t) \leq 0. \tag{19}$$

Because $\sigma_p > 0$, inequality (19) can be written as

$$\frac{1}{2}(1+\epsilon\nu_p)d_p e_p^T\Gamma e_p + \sum_{q=1}^{N}(1+\epsilon\mu_p)a_{pq}e_p^T\Gamma e_p - \frac{1}{4}\sum_{q=1}^{N}a_{pq}\tilde{z}_{pq}^T\Gamma\tilde{z}_{pq} - \frac{1}{2}d_p\tilde{\psi}_p^T\Gamma\tilde{\psi}_p \leq \frac{1}{\sigma_p}\delta_p(t). \tag{20}$$

Combining (19) with (20) yields

$$\dot{\delta}_p(t) \geq -\frac{1}{\sigma_p}\delta_p(t) - \eta\delta_p(t). \tag{21}$$

By using the comparison lemma in [39], one has $\delta_p(t) \geq 0, \forall t \in [0, \infty), p \in \mathcal{Y}$. Note that the similar methodology has also been used in [40].

Theorem 1. *Consider the leader-following multiagent system consisting of (1) and (2). We assume that the Assumptions 1–2 hold true. Then, the consensus of such a multiagent system can be reached asymptotically under the presented event-triggered control law (5). Moreover, the exclusion of Zeno behaviour can be guaranteed effectively.*

Proof. For stability analysis, the Lyapunov function candidate is chosen as

$$V(t) = \frac{1}{2}\psi^T(I_N \otimes \mathbb{F})\psi + \sum_{p=1}^{N}\left[\frac{(\mu_p - \pi)^2}{8k_p} + \frac{(\nu_p - \pi)^2}{4k_p}\right] + \frac{\pi}{2}\sum_{p=1}^{N}\delta_p(t) \tag{22}$$

where π is a positive scalar that will be defined later.

Now, we will investigate the derivative of (22) (along the trajectory of the resulting closed-loop system).

$$\begin{aligned}\dot{V}(t) = {}& \psi^T(I_N \otimes \mathbb{F}A)\psi - \psi^T(\nu D \otimes \Gamma)\tilde{\psi} - \psi^T(\mu L \otimes \Gamma)\tilde{\psi} + \sum_{p=1}^{N}\frac{(\nu_p - \pi)}{2k_p}\dot{\nu}_p \\ &+ \sum_{p=1}^{N}\frac{(\mu_p - \pi)}{4k_p}\dot{\mu}_p + \frac{\pi}{2}\sum_{p=1}^{N}\dot{\delta}_p(t),\end{aligned} \tag{23}$$

where $\tilde{\psi} = [\tilde{\psi}_1^T, \cdots, \tilde{\psi}_N^T]^T$. It is not difficult to check that the following equations and inequalities are satisfied.

$$-\psi^T(\nu D \otimes \Gamma)\tilde{\psi} = -\tilde{\psi}(\nu D \otimes \Gamma)\tilde{\psi} + e^T(\nu D \otimes \Gamma)\tilde{\psi}, \tag{24}$$

$$e^T(\nu D \otimes \Gamma)\tilde{\psi} \leq \frac{1}{2}\tilde{\psi}^T(\nu D \otimes \Gamma)\tilde{\psi} + \frac{1}{2}e^T(\nu D \otimes \Gamma)e, \tag{25}$$

$$-\psi^T(\mu L \otimes \Gamma)\tilde{\psi} = -\tilde{\psi}(\mu L \otimes \Gamma)\tilde{\psi} + e^T(\mu L \otimes \Gamma)\tilde{\psi}, \tag{26}$$

and

$$e^T(\mu L \otimes \Gamma)\tilde{\psi} \leq \frac{1}{2}\tilde{\psi}(\mu L \otimes \Gamma)\tilde{\psi} + \frac{1}{2}e^T(\mu L \otimes \Gamma)e. \tag{27}$$

Substituting (24)-(27) and $\dot{\nu}_p(t)$ into (23), one has

$$\begin{aligned}\dot{V}(t) \leq {}& \psi^T(I_N \otimes \mathbb{F}A)\psi - \frac{1}{2}\tilde{\psi}^T(\nu D \otimes \Gamma)\tilde{\psi} + \frac{1}{2}e^T(\nu D \otimes \Gamma)e - \frac{1}{2}\tilde{\psi}^T(\mu L \otimes \Gamma)\tilde{\psi} \\ &+ \frac{1}{2}e^T(\mu L \otimes \Gamma)e + \frac{1}{2}\tilde{\psi}^T(\nu D \otimes \Gamma)\tilde{\psi} - \frac{\pi}{2}\tilde{\psi}^T(D \otimes \Gamma)\tilde{\psi} \\ &+ \sum_{p=1}^{N}\frac{(\mu_p - \pi)}{4k_p}\dot{\mu}_p + \frac{\pi}{p}\sum_{p=1}^{N}\dot{\delta}_p(t).\end{aligned} \tag{28}$$

After that, substituting $\dot{\mu}_p(t)$ into (28), one has

$$\dot{V}(t) \leq \psi^T(I_N \otimes \mathbb{F}A)\psi + \frac{1}{2}e^T(\nu D \otimes \Gamma)e - \frac{1}{2}\tilde{\psi}^T(\mu L \otimes \Gamma)\tilde{\psi} + \frac{1}{2}e^T(\mu L \otimes \Gamma)e \\ - \frac{\pi}{2}\tilde{\psi}^T(D \otimes \Gamma)\tilde{\psi} + \frac{1}{2}\tilde{\psi}^T(\mu L \otimes \Gamma)\tilde{\psi} - \frac{\pi}{2}\tilde{\psi}^T(L \otimes \Gamma)\tilde{\psi} + \frac{\pi}{2}\sum_{p=1}^{N}\dot{\delta}_p(t), \tag{29}$$

which further indicates that

$$\dot{V}(t) \leq \psi^T(I_N \otimes \mathbb{F}A)\psi + \frac{1}{2}e^T(\nu D \otimes \Gamma)e + \frac{1}{2}e^T(\mu L \otimes \Gamma)e - \frac{\pi}{2}\tilde{\psi}^T(D \otimes \Gamma)\tilde{\psi} \\ - \frac{\pi}{2}\tilde{\psi}^T(L \otimes \Gamma)\tilde{\psi} + \frac{\pi}{2}\sum_{p=1}^{N}\dot{\delta}_p(t). \tag{30}$$

Note that the following expressions hold true

$$\tilde{\psi}^T(L \otimes \Gamma)\tilde{\psi} = \psi^T(L \otimes \Gamma)\psi + e^T(L \otimes \Gamma)e + 2x^T(L \otimes \Gamma)e, \tag{31}$$

$$-x^T(L \otimes \Gamma)e \leq \frac{1}{4}x^T(L \otimes \Gamma)x + e^T(L \otimes \Gamma)e. \tag{32}$$

By using $a_{pq} = a_{qp}$, we have

$$e^T(L \otimes \Gamma)e \leq \sum_{p=1}^{N}\sum_{q=1}^{N} a_{pq}e_p^T\Gamma e_p + \sum_{p=1}^{N}\sum_{q=1}^{N} a_{pq}e_q^T\Gamma e_q \\ = 2\sum_{p=1}^{N}\sum_{q=1}^{N} a_{pq}e_p^T\Gamma e_p. \tag{33}$$

Substituting (31)-(33) into (30) yields

$$\dot{V} \leq \frac{1}{2}\psi^T\left[I_N \otimes (\mathbb{F}A + A^T\mathbb{F}) - \frac{\pi}{4}L \otimes \Gamma\right]\psi + \frac{\pi}{2}\sum_{p=1}^{N}\left\{\frac{1}{2}(1 + \frac{2}{\epsilon\pi}\epsilon\nu_s)d_p e_p^T\Gamma e_p \right. \\ \left. + \sum_{q=1}^{N}(1 + \frac{2}{\epsilon\pi}\epsilon\mu_p)a_{pq}e_p^T\Gamma e_p - \frac{1}{4}\sum_{q=1}^{N} a_{pq}\tilde{z}_{pq}^T\Gamma\tilde{z}_{pq} - \frac{1}{2}d_p\tilde{\psi}_p^T\Gamma\tilde{\psi}_p\right\} + \frac{\pi}{2}\sum_{p=1}^{N}\dot{\delta}_p(t). \tag{34}$$

By substituting (18) into (34) and choosing an appropriate π such that $\pi \geq \max\{\frac{2}{\epsilon}, \frac{4}{\lambda_2(L)}\}$, where $\lambda_2(L)$ is the second smallest eigenvalue of L, one has

$$\dot{V} \leq \frac{1}{2}\psi^T\left[I_N \otimes (\mathbb{F}A + A^T\mathbb{F}) - \frac{\pi}{4}L \otimes \Gamma\right]\psi - \frac{\pi\eta}{2}\sum_{p=1}^{N}\delta_p(t).$$

Then, according to the ARE (6), we can finally conclude that

$$\dot{V} \leq -\frac{1}{2}\lambda_{\min}(Q)\psi^T\psi - \frac{\pi\eta}{2}\sum_{p=1}^{N}\delta_p(t) \tag{35}$$

Therefore, $\dot{V}(t) \leq 0$. It is further can be checked that $\dot{V}(t) = 0$ only when $\psi(t) = 0$. By using the Barbalat Lemma in [39], it follows that $\psi(t) \to 0$ as $t \to \infty$. So, consensus is reached asymptotically. Moreover, it is concluded that the signals μ_p, ν_p are bounded.

Note that although the determination of π depends on the second smallest eigenvalue of the Laplacian matrix, π is only used in the proof of consensus and is not used in neither the control protocol nor the triggering rule. Therefore, the given control method here is a fully distributed one.

In what follows, we will demonstrate that under the designed event-triggered mechanism, the Zeno behaviour can be avoided. Firstly, according to (21), we can easily get

$$\delta_p(t) \geq \delta_p(t_0) \exp\left\{-\left(\eta + \frac{1}{\sigma_p}\right)(t - t_0)\right\}. \tag{36}$$

Secondly, consider the evolution of $e_p(t)$ for $t \in [t_s^p, t_{s+1}^p)$, $t_{s+1}^p < \infty$. We have that

$$\begin{aligned} \dot{e}_p &= Ae_p - BKw_p \\ &= Ae_p + BK\mu_p \sum_{q \in N_p} a_{pq}\tilde{z}_{pq} + BK\nu_p d_p \tilde{\psi}_p. \end{aligned} \tag{37}$$

Similar to the method in [26], the time derivative of $\|e_s(t)\|$ over time interval $[t_k^s, t_{k+1}^s)$ can be written as

$$\frac{d\|e_p(t)\|}{dt} = \frac{e_p^T}{\|e_p\|}\dot{e}_p \leq \|\dot{e}_p\| \leq \|A\|\|e_p\| + \mu_p \sum_{q \in \mathcal{N}_p} a_{pq}\|BK\|\|\tilde{z}_{pq}\| + \nu_p d_p \|BK\|\|\tilde{\psi}\|. \tag{38}$$

As shown above, both μ_p, ν_p and ψ are bounded. The last of which indicates $x_p(t) - x_q(t), \forall (p, q) \in \mathcal{Z}$, is bounded. Generally, assume that $\mu_p \leq \bar{\mu}$ and $\nu_p \leq \bar{\nu}$ for some positive constants $\bar{\mu}$ and $\bar{\nu}$. It is not difficult to get that the interval time between two consecutive triggering instants is bounded. Hence, $e^{A(t - t_s^p)}$ is also bounded for any $t \in [t_s^p, t_{s+1}^p)$. Due to $x_p = \psi_p + x_0$, we can easily get that $x(t)$ is limited for any limited t. Therefore, $\tilde{z}_{pq}$ is also bounded. So, we can derive from (38) that

$$\frac{d\|e_p(t)\|}{dt} \leq \|A\|\|e_p(t)\| + \bar{\mu}\varrho_p + \bar{\nu}\gamma_p, \tag{39}$$

where ϱ_p is defined as the upper bound of $\sum_{l \in N_p} a_{pq}\|BK\|\|\tilde{z}_{pq}\|$ and γ_p denotes that of $d_p\|BK\|\|\tilde{\psi}_p\|$. Consider a function ϕ which is non-negative and satisfies the following initial condition

$$\dot{\phi} = \|A\|\phi + \bar{\mu}\varrho_p + \bar{\nu}\gamma_p, \phi(0) = \|e_p(t_s^p)\| = 0. \tag{40}$$

After that, we can get that $\|e_p(t)\| \leq \phi(t - t_k^p)$, where $\phi(t)$ is the solution to (40), shown as

$$\phi(t) = \left[(\bar{\mu}\varrho_p + \bar{\nu}\gamma_p)/\|A\|\right](e^{\|A\|t} - 1). \tag{41}$$

Assume that initial time $t_0 = 0$. If we have the following condition, then it is to check that triggering function (15) satisfies $f_p(t) \leq 0$,

$$\|e_p\|^2 \leq \frac{\delta_p(t_0) \exp\left\{-\left(\eta + \frac{1}{\sigma_p}\right)(t)\right\}}{\frac{1}{2}d_p(1 + \epsilon\bar{\nu})\|K\|^2 + d_p'(1 + \epsilon\bar{\mu})\|K\|^2}. \tag{42}$$

Therefore, a lower bound Δ of $t^p_{s+1} - t^p_s$ satisfies:

$$\frac{(\bar{\mu}\varrho_p + \bar{\nu}\gamma_p)^2}{\|A\|^2}(e^{\|A\|\Delta} - 1)^2 \geq \frac{\delta_p(t_0)\exp\left\{-\left(\eta + \frac{1}{\sigma_p}\right)\left(t^p_s + \Delta\right)\right\}}{\frac{1}{2}d_p(1+\epsilon\bar{\nu})\|K\|^2 + d'_p(1+\epsilon\bar{\mu})\|K\|^2}. \tag{43}$$

Then, we get that

$$t^p_{s+1} - t^p_s \geq \Delta \geq \frac{1}{\|A\|}\ln\left[1 + \frac{\|A\|}{\|K\|(\bar{\mu}\varrho_p + \bar{\nu}\gamma_p)}\sqrt{\frac{\delta_p(t_0)\exp\left\{-\left(\eta + \frac{1}{\sigma_p}\right)\left(t^p_s + \Delta\right)\right\}}{\frac{d_p}{2}(1+\epsilon\bar{\nu}) + d'_p(1+\epsilon\bar{\mu})}}\right] \tag{44}$$

Note that the lower bound of Δ proposed in (44) closes to zero only when $t \to \infty$. Then, we can conclude that Δ always exists and is strictly positive in any limited time. Therefore, the Zeno behaviour is excluded. □

Remark 1. There are plenty of pioneer results concerning on event-based cooperative control of multiagent systems. However, most of them depend on some global information of the whole network, for example, the smallest non-zero eigenvalue of Laplacian matrix in [24] and number of agents in [25,41,42] are used. Such information is usually inaccessible for each agent, and thus the obtained results are somewhat conservative in terms of the usability. However, in this paper, the developed strategy relaxes the requirement of the global information and thus is more general compared with some existing results.

Remark 2. Note that there also have some existing results concerning on the fully distributed event-based controller, such as [26,27,29]. In [27], the leaderless consensus is investigated, and the obtained results cannot address the leader-following consensus successfully. In [26], the adaptive coupling weights are edge-based, which limits its expansion especially when the communication structures are complex. In [43], the bipartite leader-following consensus is studied, which is the different control task compared with this paper. Moreover, all results mentioned above are presented in traditional static event-based protocols. However, we investigate a dynamic event-triggered control design problem so that the sampling times of each followers can be reduced due to the introduction of the auxiliary internal variable $\delta_p(t)$. It should be pointed out that the dynamic variable $\delta_p(t)$ is always non-negative, which has been proved by (18)-(21). Note that $\delta_p(t)$ is a term of $f_p(t)$ with a negative sign and the new events will be generated when $f_p(t) \geq 0$ according to (10). Therefore, the introduction of $\delta_p(t)$ makes it more difficult to generate the next events by raising the trigger threshold. And the number of triggers is reduced compared to the case without such dynamic parameters (similar to existing results, we called it as the static event-triggered rule). Then the limited communication bandwidth can be saved via our approach. The main challenges and difficulties of our works come from the design of the novel triggering rule under fully distributed framework.

Remark 3. Compared to the existing fully distributed event-triggered result in [26,36], the node-based adaptive parameters $\mu_p(t)$ and $\nu_p(t)$ are developed in this paper. Such a design can effectively avoid the complexity of the controller in the large-scale network. Moreover, the edge-based adaptive parameter $c_{ij}(t)$ is required to satisfy $c_{ij}(t) = c_{ji}(t)$. The equation is easy to be destroyed by communication noises in the practical systems, and thus the control performance will be deteriorated. The node-based adaptive parameters $\mu_p(t)$ and $\nu_p(t)$ in this article removes this limitation.

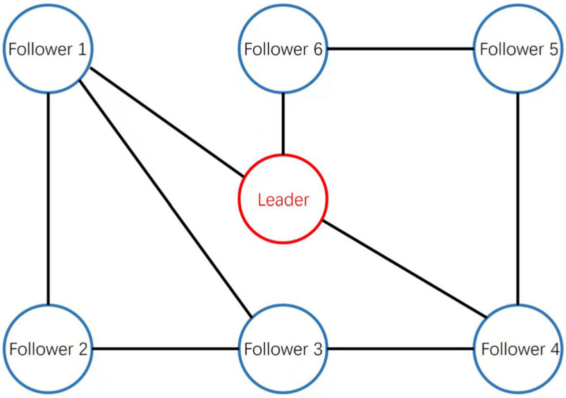

Fig. 1. Topology structure of the studied multiagent system.

Remark 4. In [36], the authors investigate the leaderless consensus under fully distributed dynamic event-triggered control. However, such a control protocol cannot be extended to the leader-following case. In the leaderless case, the coefficient of $\delta_s(t)$ in the (35) can be independent from the π by using some inequalities. But the additional terms caused by the introduction of the leader agent like $\frac{1}{2}e^T(\nu D\otimes\Gamma)e$ and $\frac{\pi}{2}\tilde{\psi}^T(D\otimes\Gamma)\tilde{\psi}$ in (30) cannot be scaled by the inequalities provided in [36]. Therefore, we have to find another way to relax the limitation that the dynamic variable should use some global information to guarantee the leader-following consensus. The dynamic event-triggered consensus control is also investigated in [37]. Though the authors claim that their control protocol is fully distributed, some global information has to be known before. In the paper, the decreasing rate η of the dynamic variable $\delta_p(t)$ should satisfy $\eta\geq\frac{2}{\lambda_2(L)}$, which means we have to know the Laplacian matrix of the communication network in advance to calculate and design a suitable η. In a short, the introduced dynamic variable still involves some global information to guarantee the convergence of the Lyapunov function candidate. Such a problem also brings challenge to us, and it becomes more difficult in the leader-following case. To resolve this problem, we develop a novel Lyapunov function candidate $V(t)$ in wich an additional term $\frac{\pi}{2}\sum_{p=1}^{N}\delta_p(t)$ is introduced, and any positive constant η can ensure the stability of $V(t)$. The global information in the dynamic variable $\delta_p(t)$ is excluded.

4. Simulation examples

In this section, a group of unmanned intelligent vehicles are provided to demonstrate the validity of the given method.

In this example, a leader and six followers are considered, the communication graph among them are given in Fig. 1. According to [44] and [45], each intelligent vehicle can be modeled by

$$\begin{cases}\dot{C}_i^x(t)=v_i(t)\cos(\Theta_i(t))\\ \dot{C}_i^y(t)=v_i(t)\sin(\Theta_i(t))\\ \dot{\Theta}_i(t)=\omega_i(t)\end{cases}\tag{45}$$

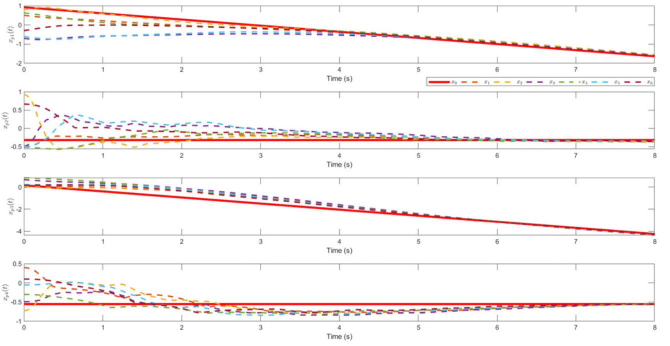

Fig. 2. The state trajectories of agents.

where $C_i^x(t)$ and $C_i^y(t)$ are the horizontal and vertical coordinates of vehicles' centroid, respectively. $\Theta_i(t)$ is the heading angle; $v_i(t)$ is the linear velocity and $\omega_i(t)$ is angular velocity. By applying the dynamic feedback linearization method described in [44], unmanned intelligent vehicle model (45) can be transformed into (1) and (2) with

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Other controller parameters are chosen as $k_p = 0.2$, $\eta = 1$, $\epsilon = 1$, $\sigma_p = 1$, $\delta(t_0) = 1$ and Q is selected as an identity matrix I_4. In this simulation, the initial values of the agents are randomly selected between -1 and 1. It is not difficult to verify that Assumption 1 and Assumption 2 are both satisfied for this multiple unmanned intelligent vehicles system. After that, by solving the ARE (6), we can obtain a positive definite matrix $\mathbb{F}$ as

$$\mathbb{F} = \begin{bmatrix} 1.7321 & 1 & 0 & 0 \\ 1 & 1.7321 & 0 & 0 \\ 0 & 0 & 1.7321 & 1 \\ 0 & 0 & 1 & 1.7321 \end{bmatrix}. \tag{46}$$

Then, we have

$$K = \begin{pmatrix} 1 & 1.7321 & 0 & 0 \\ 0 & 0 & 1 & 1.7321 \end{pmatrix}.$$

By using the designed event-based control law in previous section, the states of each agent are shown in Fig. 2, and the event-triggered instants of each agents are presented in Fig. 3. The triggering numbers of 6 followers are 12,46,37,11,28 and 12, respectively. From these figures, we can conclude that that the state trajectories of the six unmanned intelligent vehicle systems can follow the leader's value successfully, and the Zeno phenomenon of the system can be avoided. Moreover, the variation tendencies of $\mu_p(t)$ and $\nu_p(t)$, $p = 1, \ldots, 6$ are depicted in Fig. 4 and Fig. 5, respectively. It is easy to check that ν_2, ν_3 and ν_5 are always zero because these followers have no connection with the leader agent. Other adaptive coupling weights monotonically increase and converge to some positive constants, which meets our expectation. Finally, the non-negative dynamic internal variables $\delta_p(t)$ are presented in Fig. 6.

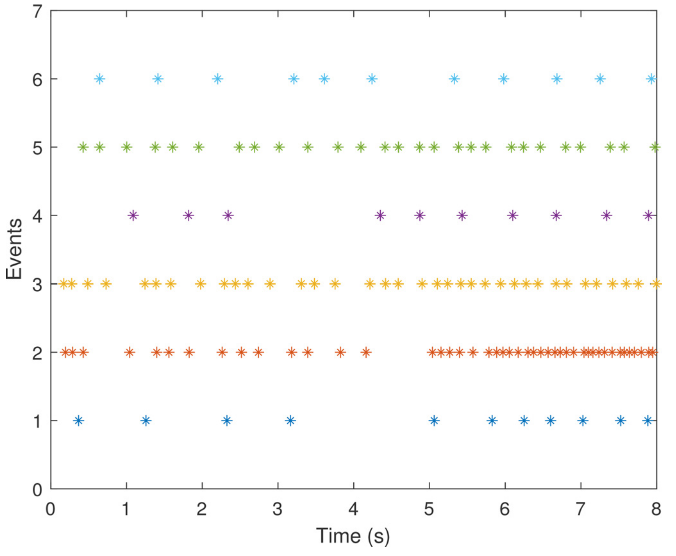

Fig. 3. Triggering instants of 6 followers.

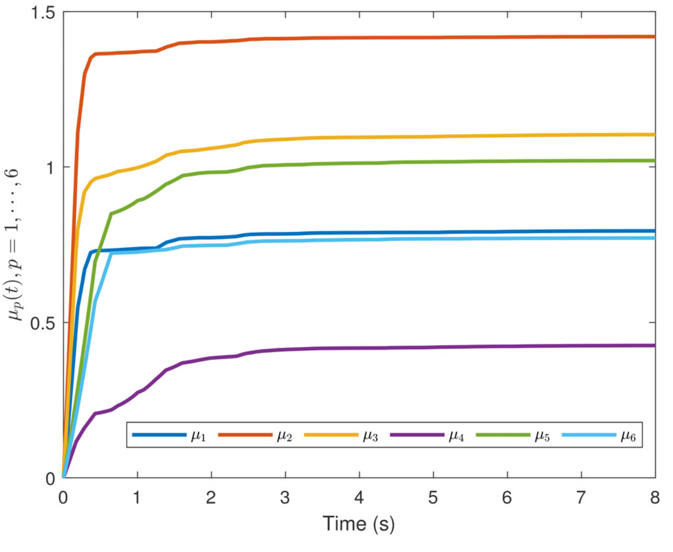

Fig. 4. Adaptive coupling weights $\mu_p(t)$.

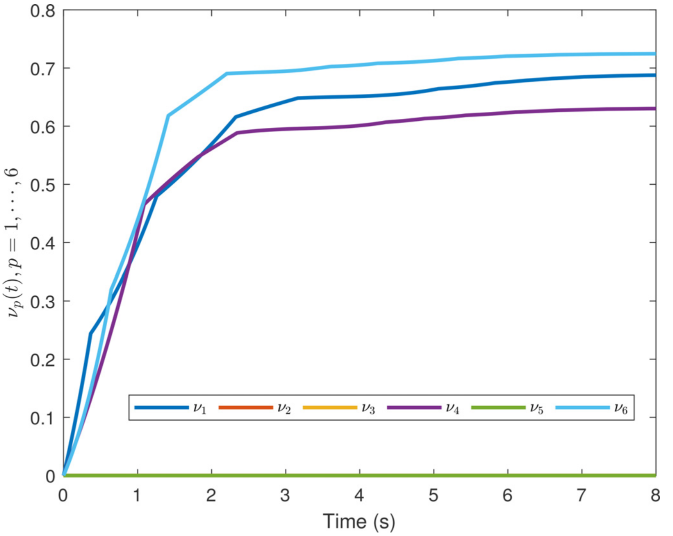

Fig. 5. Adaptive coupling weights $\nu_p(t)$.

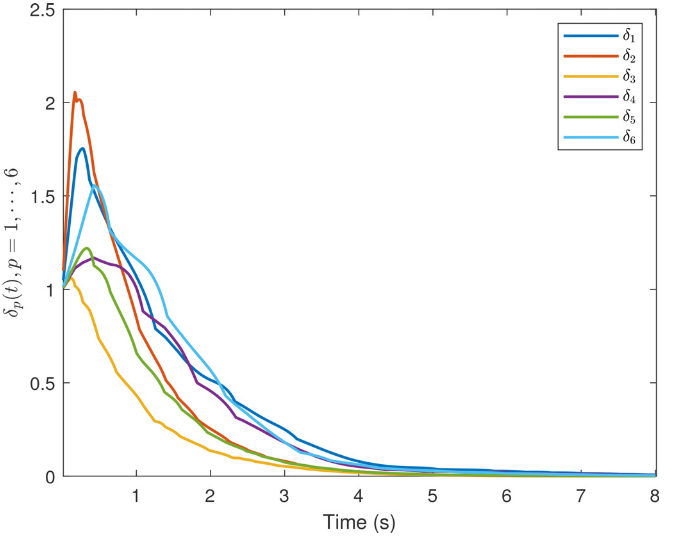

Fig. 6. Dynamic internal variables $\delta_p(t)$.

5. Conclusion

In this article, the fully distributed dynamic event-based controller has been designed to address leader-following consensus problem of multiagent systems with general dynamics. By designing some adaptive variables and introducing an internal variable for each agent, a fully distributed event-triggered mechanism without any requirement of global information has been developed. It has been proved that the designed fully distributed protocol can resolve the leader-following consensus problem successfully, moreover, the Zeno behaviour are proved to be excluded efficiently. A group of unmanned intelligent vehicle systems have been proposed to prove the feasibility and the validity of the designed control mechanism. However, the control protocol that we desigend is generally complicated. How to designed a more simple and efficient protocol is one of our future tasks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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